Lohith Burra

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EU Chancenkarte visa - authorized to work in Germany/EU immediately, no sponsorship required

SUMMARY

Automation engineer who turns manual, judgment-heavy engineering work into reusable Python packages with tests, validation and documentation, built to run unattended and to fail loudly rather than silently. Writes the code, and enables non-specialists to write their own. Mechanical engineering background, MS in Data Science, hands-on RAG and agent stack.

CORE SKILLS

Python: packages, code generators, FastAPI, CLI tooling, automated test suites, JSON-schema validation

AI & Retrieval: Claude API, Gemini, LangGraph, MCP tools, RAG, pgvector, local embeddings, LLM benchmarking

Automation: n8n (23+ workflows), workflow generators, GitHub Actions CI/CD, REST APIs, idempotency design

Infra & Data: Docker, Kubernetes, PostgreSQL, TypeScript, Apache Spark, PySpark, ETL, AWS (S3, Lambda, Glue)

PROFESSIONAL EXPERIENCE

Manager, AI & Automation Jun 2026 - Sep 2026

Corominas Consulting - Remote, Germany

- Owned the automation layer across roughly 85 deployments: built the shared Python packages each one composes, with per-deployment behaviour in configuration rather than code branches, so a new environment is a config change and not a rebuild
- Generated orchestration instead of hand-building it: 26 workflow definitions, 490 nodes, the production ones emitted by Python generators, so a policy change is a reviewable diff of a generator rather than of a 48 KB JSON blob
- Wrote the deterministic validation layer that makes automated output safe to ship unreviewed: 18 checks across 10 families, backed by 715 assertion calls across 33 test scripts
- Engineered for unattended operation: idempotency before retries, 3 tries with a 300-second timeout across 137 service-facing calls, every destructive path dry-run by default with snapshot and revert
- Enabled a non-specialist colleague to ship their own automation: the domain lead authored the domain logic, I built the runtime that executes it, then pushed automation inside those units so each self-encapsulates

AI Automation Developer Aug 2025 - Jun 2026

Formula Dream - Remote | Contract

- Built an unattended pipeline end to end: GitHub Actions running self-scheduling Python render jobs into a 21-node orchestration workflow with dual triggers, idempotent deduplication, retry with backoff, and failure mail naming the exact node
- Built a meta-tool so others could build too: Claude generates complete importable workflow definitions pre-configured for the target credential setup, idea to running workflow in under 10 minutes. 23+ workflows built

AI Engineer Apr 2025 - Mar 2026

Think Round Inc - Remote, US

- Built a production conversational AI system with memory persistence, confidence scoring and multi-layered deterministic safety controls, on a multi-agent architecture supporting parallel development of independent agents on shared infrastructure

Program Analyst Feb 2022 - Aug 2023

Cognizant Technology Solutions - India

- Built Python and Apache Spark ingestion pipelines processing 50M+ rows daily, cut data-to-KPI latency from 48 hours to 2 hours, and optimised ETL on AWS (S3, Lambda, Glue) with measurable cost reduction

KEY PROJECTS

SceneAgent - Queryable 3D Scene Intelligence Apr 2026 | github.com/lohithburra01/sceneagent

- Built in 48 hours. Point cloud, photos or BIM in, semantically queryable scene out: segmentation, backprojection and DBSCAN clustering resolved 151 distinct objects
- Geometry exposed as atomic tools an agent calls over MCP: locate by natural language, measure distances, plan a route. FastAPI, pgvector, Docker, Kubernetes

EDUCATION

Master of Science - Data Science - Stevens Institute of Technology, USA (2025)

Bachelor of Technology - Mechanical Engineering - JNTU K, India (2022)